

KIMBERLY MCNEIL

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EDUCATION

Muhlenberg College, Allentown, PA

Expected Graduation May 2026

Bachelor of Science in Computer Science; Minor in Mathematics

GPA: 3.968, Dean's List (2022–Present), Competitive research grant recipient

Relevant Courses: Computer Organization, Data Structures, Database Systems, Software Engineering, Introduction to Machine Learning, Statistical Models, Probability, Human-AI Interaction, Data Wrangling, Spatial Data Mining

PROFESSIONAL EXPERIENCE & LEADERSHIP

Academic Resource Center (Workshop) Tutor, Muhlenberg College, Allentown, PA **September 2023 – Present**

Scholar

- Met weekly with individuals or groups of students for introductory and intermediate computer science courses, mainly teaching Python, Java, and Data Structures and Algorithms.
- One of two selected to facilitate the General Computer Science Workshop, where we assisted students with debugging code, understanding coding logic, and building confidence for exams.

Geospatial Data Science Portfolio, Allentown, PA

January 2026 – May 2026

Scholar, Researcher

- Manhattan Cycling & Income Story Map (QGIS, qgis2web): Built an interactive browser-based GIS map joining Opportunity Atlas income data to 2020 census tracts across 6 spatial layers; identified a positive correlation between higher-income Manhattan neighborhoods and bike infrastructure density.
- NYC Unemployment Spatial Autocorrelation (Python, GeoPandas, PySAL): Applied Global Moran's I and LISA cluster analysis to 2023 unemployment data across all 62 NYS counties; found statistically significant spatial clustering ($I = 0.13$, $p = 0.012$) concentrated in southern New York.
- GeoAI Building Detection Model Comparison (Python, PyTorch, Rasterio): Trained and benchmarked U-Net, DeepLabV3, and SegFormer on aerial imagery for building footprint detection; U-Net achieved the strongest results ($IoU = 0.727$, $F1 = 0.826$).

Quantum Machine Learning Research, Muhlenberg College, Allentown, PA

June 2025 – August 2025

Scholar, Researcher

- Analyzed and presented the performance metrics of classical Support Vector Classifier (SVC) and Quantum-Hybrid Variational Quantum Classifier (VQC), providing insights into the performance differences between classical and quantum approaches for classification.
- Leveraged Python libraries including scikit-learn, Qiskit, pandas, seaborn, matplotlib, numpy, and pympler to effectively preprocess data, implement machine learning models, and visualize results.

ADDITIONAL ACTIVITIES

Activities Council, President, Muhlenberg College

September 2023 – Present

- Led a team of ~15 students, and collaborated with other executive team members, artists, and companies to create, plan, and facilitate over 200 unique and engaging events and programs in a collegiate setting with a strong and steady attendance.
- Implemented new organizational strategies and documentation after shifts in institution management due to the pandemic shutdown, improving accessibility and reducing search times by 50%.

Computer Science Society, Participant, Muhlenberg College

October 2022 – Present

SKILLS

Computer: Google Workspaces, Python, R, SQL, HTML, data management.

Other: documentation, grant writing, event planning, organization, research, writing